Can we 'stop' global warming?

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August 9, 2026

Abstract

This report produces a simple mathematical model to investigate whether climate change can be 'stopped' by altering economic growth, technology, and implementing active carbon removal strategies. The model simplifies the Kaya Identity and combines the expression with a linear relationship between carbon emissions and temperature, as well as a model which represents the effects of temperature on sea levels. This report examines the mathematical and practical feasibility of three GICCE climate goals. Results indicate that improvements in technology and reduced economic growth can substantially slow warming, but stabilising temperatures requires net emissions to approach zero, and reducing temperatures requires sustained net-negative emissions. The most ambitious goal requires carbon removal to exceed 90 gigatonnes annually by the end of the century. All scenarios for achieving climate stabilisation are mathematically achievable within the model, however, few are likely to be practically feasible under current technological and economic conditions.

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1 Introduction

Climate change is one of the most significant global environmental challenges, driven largely by anthropogenic greenhouse gas emissions from fossil fuel combustion, industrial processes and land-use changes [1]. The resulting carbon dioxide (CO_2) emissions into the atmosphere has caused a sustained rise in global mean temperature, leading to widespread environmental challenges. As part of the initiative of the Global Institute for Climate Change and the Environment (GICCE), this report seeks to improve general understanding of the Earth's climate and develop a simple mathematical model which reflects the goal of international governments to 'stop' climate change.

Human activities are the primary driver of increasing atmospheric CO_2 [1]. Fossil fuel combustion, land-use changes and other industrial processes release large quantities of carbon which exceed the Earth's natural ability to absorb it through carbon sinks [2]. The Kaya Identity provides an expression for determining annual CO_2 emissions. The identity decomposes CO_2 into a product of population, GDP per capita, energy intensity and carbon intensity. Deutch (2017)[3] uses this expression to examine the relationship between economic growth and emissions, and demonstrates that improvements in energy and carbon intensity can weaken this relationship. This is important for the purposes of the current report as it allows a framework for proposing technological changes to produce a reduction in annual emissions. The "Global Carbon Budget 2025" (GCB) provides empirical critiques against the assumptions of the model. Friedlingstein et al. (2026) states that global emissions have not yet reached a sustained decline despite energy and carbon intensity improvements, which is why global anthropogenic CO_2 emissions have continued to increase by 0.3% per year in the decade 2015–2024. However, it also identifies evidence of partial decoupling. This report aims to adopt the relationship established by the Kaya Identity, but it will assume that technological advancements have a direct impact on energy and carbon intensity. Our model must also be closely examined against observed trends.

Matthews et al. (2009)[4] found a linearly proportional relationship between global warming and cumulative carbon emissions from semi-empirical sources. The constant of proportionality is the Climate-Carbon Response (CCR), which is estimated to be approximately 1–2.1°C of warming per trillion tonnes of carbon (TtC) emitted. This makes cumulative emissions highly useful for examining scenarios in which temperature is required to stabilise. However, Matthews et al. calculated substantial uncertainty on the model. This is due to the various assumptions on which the relationship relies, including that other greenhouse gases and non-greenhouse gases contribute negligibly to global warming, and that historic land-use changes and aerosol forcing from limited available data is accurate. Uncertainty in this data would have great effects on the CCR value. The authors also note that their results are derived from idealised climate-carbon model scenarios over decades to centuries, meaning that extrapolation over longer time scales may not produce accurate results. The model produced in this report will only rely on Matthews et al.'s linear relationship, and will estimate a constant of proportionality which is solely based on empirical data.

Population represents a fundamental driver of emissions due to rising demands for energy. Lee (2003)[5] summarised the demographic transition theory, which describes the growth changes of a population as a region develops. The paper is helpful to produce a logistics model for long-term demographic projections. However, the literature identifies several factors which would affect population trajectories, suggesting that a carrying capacity in a simple logistics model may not accurately represent the social, economic or demographic mechanisms which would affect future population growth. Our logistic model is therefore used as an approximation for long-term behaviour and assumes that such behaviour is stable over long timescales.

Stabilising global temperature does not capture all risks associated with climate change. Sea-level rise continues long after temperatures stabilise due to delayed responses of ocean heat uptake and ice-sheet dynamics. Rahmstorf (2007)[6] proposed a semi-empirical linear relationship between sea-level changes and temperature. The model captures long-term sea level behaviour as opposed to an assortment of underlying physical processes. While the model performs well over longer timescales, it is less reliable over shorter periods as it does not account for fluctuations in physical processes. Nonetheless, it provides an appropriate framework for assessing whether sea-levels can eventually stabilise under changing climate scenarios.

The literature provides key relationships required for the model in this report, which will be applied to three scenarios by the GICCE. The first is stabilisation of global temperatures to 4°C above pre-industrial levels by 2100. The second is stabilisation at 2°C by 2050, and the third is limiting warming to no higher than 1.5°C while plateauing sea-level rise by 2100. The model is subsequently used to assess the various changes and advancements required to achieve each scenario and evaluate the implications for the environment and humanity.

2 Modelling Methodology

2.1 Main Models

To model temperature change, we modified the Transient Climate Response to Cumulative Carbon Emissions (TCRE) relationship[4]. Matthews et al. established a linear relationship between Global Surface Temperature anomaly and Cumulative Carbon Emissions, both since pre industrial levels, which we model as

$$\Delta T(t) \propto \int_{t_0}^{t} E(y)dy, \tag{1}$$

where:

- t is time in years;
- t_0 is the year we start measuring from;
- $\Delta T(t)$ is the temperature change at time t since t_0; and
- $E(t)$ is the amount of CO_2 produced by humans in year t measured in tonnes.

Note that emissions and CO_2 emissions are used interchangeably, as CO_2 accounts for a majority of the global heating due to greenhouse gas emissions, and a majority of greenhouse gases emitted [7].

We modify this by introducing a reduction term that subtracts from the emissions, representing technological efforts to directly remove carbon from the atmosphere. This leads to a new equation, which is the primary model for the paper

$$\Delta T(t) \propto \int_{t_0}^{t} (E(y) - R(y))dy, \tag{2}$$

where R is a function representing annual carbon reduction efforts. To model anthropogenic carbon emissions E, we used the Kaya Identity[3], which states

$$E(t) \propto \text{Population} \times \frac{\text{GDP}}{\text{Population}} \times \frac{\text{Energy used}}{\text{GDP}} \times \frac{\text{CO}_2 \text{ produced}}{\text{Energy used}}.$$

In other words, CO_2 emissions is the product of population, GDP per capita, energy intensity, and carbon intensity. To simplify our model, we capture the product of the latter two as technological inefficiency D in terms of GDP per tonne of CO_2. This gives a simplified Kaya Identity

$$E \propto PYD, \tag{3}$$

where:

- P is population;
- Y denotes global average GDP per capita; and
- D is the technological inefficiency term.

2.2 Sub-Models

Next, we look at these terms in more detail.

2.2.1 Population

Population is not an aspect of the model that will be altered as a parameter in the scenarios. The first iteration of modelling this was using a logistic equation, giving population as

$$P(t) = \frac{P_{\text{final}}}{1 + (\frac{P_{\text{final}}}{P_{\text{initial}}} - 1)\exp\left(-r_p(t-1950)\right)},$$

where:

- 1950 is the start of the modelling period;
- P_{initial} is the population in 1950;
- P_{final} is a carrying capacity initially set to 10 billion, based on UN estimates[8]; and
- r_p is the rate of population growth, which has been fitted[9] to ≈ 0.031 in our model.

However, a weakness of logistic equations in modelling human population was that it did not capture realistic projections of human populations, which are projected to decline by the end of the century[8], while the logistic equation tends asymptotically to the carrying capacity.

To better model the projections of falling populations in the future, we added a second logistic equation that subtracts from the original one. This second term is referred to as the population decline term, giving us a final population model

$$P(t) = \frac{P_{\text{final}}}{1 + (\frac{P_{\text{final}}}{P_{\text{initial}}} - 1)\exp\left(-r_p(t-1950)\right)} - \frac{P_{\text{dec\_final}}}{1 + (\frac{P_{\text{dec\_final}}}{P_{\text{dec\_init}}} - 1)\exp\left(-r_{\text{dec\_p}}(t-2050)\right)}, \tag{4}$$

where:

- $P_{\text{dec\_init}}$ is set to 250 million;
- $P_{\text{dec\_fin}}$ is the amount population will decline, set to 2 billion for our model;
- $r_{\text{dec\_p}}$ is set to 0.04; and
- P_{final} is updated to 12 billion, letting the asymptotic population remain 10 billion.

2.2.2 Economy

The economic term Y, in terms of GDP per capita, is one of the parameters modelled that will be adapted to meet scenarios. It is modelled simply as an exponential, given as

$$Y(t) = 1000 + A_y e^{r_y t}, \tag{5}$$

where:

- A_y is an adjustment factor to fit historical data;
- r_y represents the rate of growth of GDP / capita; and
- \$1000 is the historical baseline GDP, which was a quirk of the data used[10].

The terms A_y and r_y were fitted to historical data[9].

In the modelling scenarios, Y is treated as a piecewise continuous function where, starting from 2027, the function's growth rate reduces to a fraction of the original growth rate.

The choice of exponential is appropriate for the period of time the model is relevant for.

While GDP is effectively related to population, as a larger population implies a rise in GDP, GDP per capita represents income per person, which is influenced more greatly by technological and productivity improvements than population.

Technological and productivity gains are in turn, derived from a sizable working age population that continually innovate and improve [11]. A sizable working age population will remain so long as there is a young workforce present in the world, and this is the case for at least until 2100, even with the diminishing population towards the end of the century [8].

2.2.3 Technological Inefficiency

This term D broadly represents the emitted CO_2 per unit of GDP. As research continues in the future, improvements in technology should result in lower carbon and energy intensities, lowering this value. It is modelled as the exponential

$$D(t) = A_d e^{r_d t}, \tag{6}$$

where:

- A_d is an adjustment factor; and
- r_d represents the rate of decreased inefficiency each year.

Both of these terms were fitted to historical data[9].

The technology term is also used as a parameter in the scenarios, where the rate of growth of technology is altered starting from the year of action (2027). For example, in scenario 1, we model technological improvements to be 1.5 times faster, directly improving energy intensity and carbon intensity 1.5x more than original projections.

2.3 Reduction Term

In Equation 2, a reduction term was introduced. This represents another technological aspect to the model, representing human efforts to remove carbon from the atmosphere. Initially, natural carbon sinks were considered to be a part of the reduction term, but the TCRE relationship[4] implicitly accounts for natural sink behaviour in its model, specifically using cumulative human emissions.

Hence, the reduction term must also denote human efforts to remove carbon from the atmosphere. Examples of initiatives include engineering solutions such as Direct Air Capture, and nature based solutions such as reforestation and regenerative agriculture.

In our scenario modelling, the reduction term is the primary driver to offsetting the carbon emissions. Constraining economic growth and improving carbon and energy intensities only reduce the amount of emissions; they do not eliminate it. Achieving net zero targets in the model requires significant carbon removal from the atmosphere, and hence the reduction term is used predominantly in the model.

When modelling scenarios, the term will use a rising function, representing increasing capacity of technology to remove carbon from the atmosphere and construction effort as time moves forward. This will be adjusted to the scenarios to fit each target.

2.4 Sea Level

Scenario 3 requires sea levels to stop rising by 2100. For this, Rahmstorf's semi empirical approach to projecting sea level rise was used. Here, Rahmstorf's model approximated the rate of change of sea levels to be proportional to the temperature anomaly from pre industrial levels to predict long term trends, which solves our purpose. Rahmstorf's model stated

$$\frac{dH}{dt} = \alpha(T - T_0), \tag{7}$$

where:

- H represents sea level;
- T is the global surface temperature;
- T_0 is the pre-industrial baseline temperature; and
- α is the proportionality constant, which was calculated[6] to be $3.4\frac{mm}{year}$ per degree Celsius.

This model simplifies by not accounting for the lag between temperature and sea levels, mainly considering long-term trends and averages [6].

3 Results

We have been given a set of scenarios to model, each with specific temperature targets and dates. So we simulate efforts to actively remove carbon from the atmosphere from 2027. In addition to the active carbon removal, we also adjust the growth rate of Y and decay rate of D after that year to find combinations of economy growth change, technological improvements, and active removal that satisfy the scenarios.

3.1 Scenario 1

For a maximum target of 4°C above pre-industrial levels by 2100, we reduced the economy growth rate r_y to 70% of its original value and increased the technology improvement rate r_d by 1.5 times.

Then we set our active removal R to linearly ramp up to 3 $GtCO_2$ per year by 2077 and continue, reaching and remaining at 5 $GtCO_2$ per year.

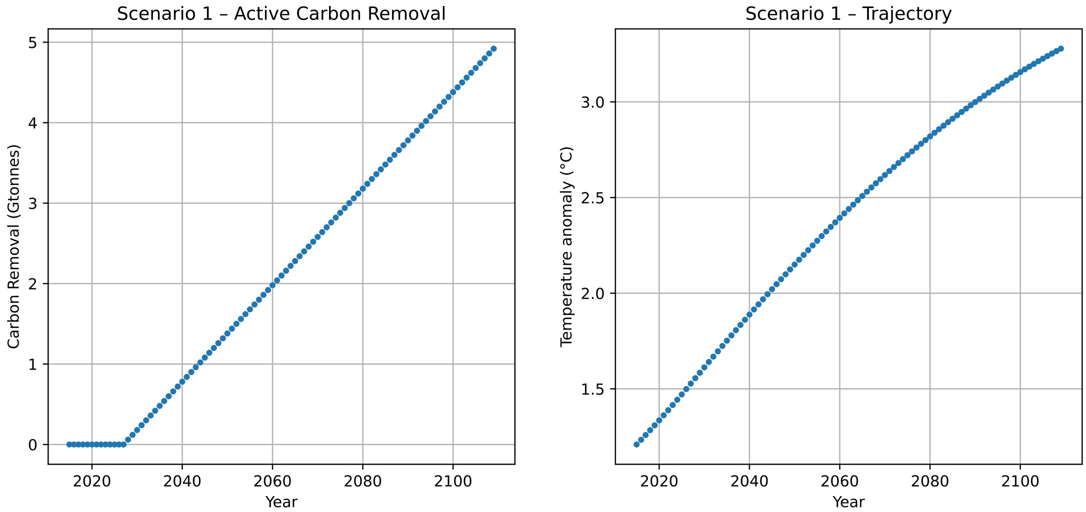

Figure 1: Scenario 1 active CO_2 removal and resulting temperature projections

So in this model, the temperature anomaly remains under 4°C until the year 2250, plateauing from there (see Appendix). Reducing the effectiveness of carbon removal by 90% causes the temperature to increase by 3.17°C in 2100, plateauing above 4.1°C after 2250.

3.2 Scenario 2

For a maximum target of 2°C above pre-industrial levels by 2050, we reduced the economy growth rate r_y to 60% of its original value and increased the technology improvement rate r_d by 2 times.

Then we set our active removal R to be a growing portion of the annual emissions. This ramps up to nearly 100% by 2050, making it plateau afterward.

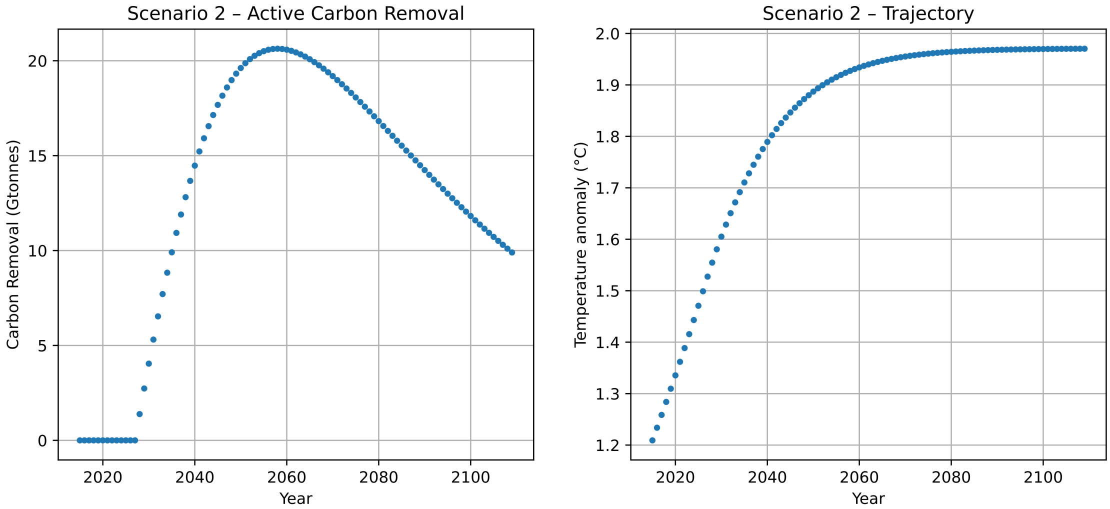

Figure 2: Scenario 2 active CO_2 removal and resulting temperature projections

So in this model, the temperature change remains under 1.9°C in 2050, and remains under 2°C. If we consider a situation where removal is only 90% of the target, the model reaches 2°C by 2070, and reaches about 2.1°C.

3.3 Scenario 3

This scenario requires a maximum temperature anomaly of 1.5°C and for sea levels to stop rising by 2100. Unfortunately, our temperature model in Figure 9 indicates that we have already surpassed the 1.5°C target as of 2025. So we only consider the sea level requirement.

From Equation 7, we must have the temperature change reach 0°C by 2100 for this to occur. So we model more aggressively, first reducing the economy growth rate r_y to 50% of its original value and increasing the technology improvement rate r_d by 2.5 times. For the active carbon removal, we follow a linear ramp, reaching 94.9 gigatonnes of CO_2 by 2100, then lowering efforts to match global emissions in order to keep temperature stable.

This results in a peak above 1.75°C before 2050, which drops to 0°C before 2100. Using the modelled temperature from Figure 3, we can determine the expected sea levels. So from Equation 7, we approximate the sea level increase since 2015

$$H(t) = \alpha \int_{2015}^{t} (T(y) - T_0) dy,$$

with results shown in Figure 4. In this model, the sea levels stop rising at 381 mm higher than in 2015.

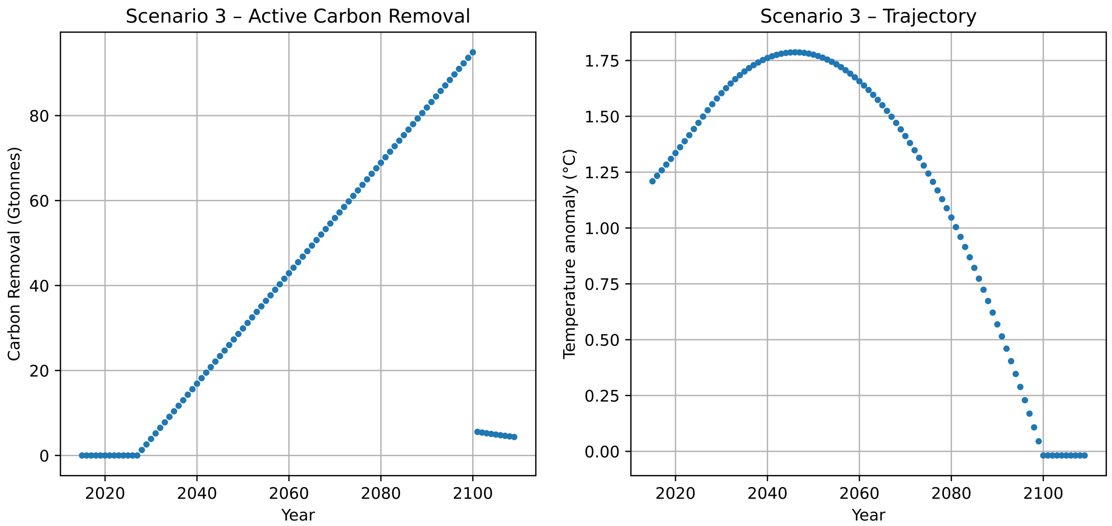

Figure 3: Scenario 3 active CO_2 removal and resulting temperature projections

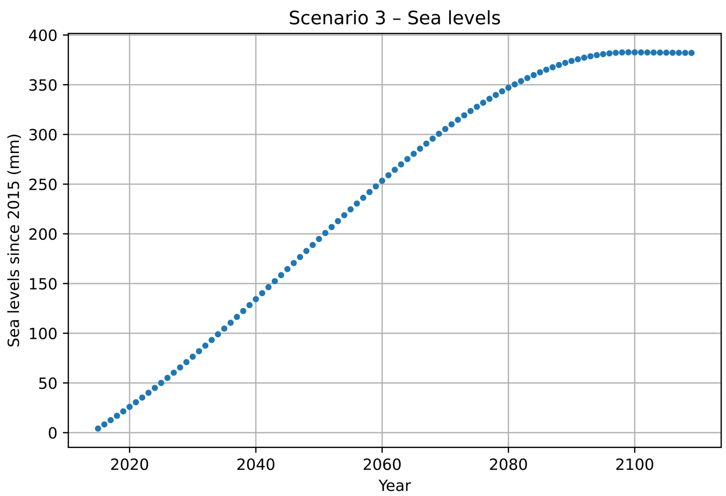

Figure 4: Scenario 3 model sea levels since 2015

If the removal efforts were only 90% as effective in this model, the linearly increasing effort to remove CO_2 would need to continue for four additional years in order to bring temperatures back to pre-industrial levels. This would also cause the projected sea level increase to 404 mm, about 23 mm higher than the original prediction.
An additional factor not considered in this model is the temperature lag for the ocean. The effects of temperature change may be reflected decades later in the ocean. Since this model would bring the temperature down before the ocean levels finished rising, the true peak sea level could be much lower than what is predicted here.

4 Discussion

Our three scenarios showcase that more ambitious climate change targets require increasingly large changes to the economic, technological and carbon-removal components of the model. In scenario 1, GDP per capita growth is reduced to 70% of the original projection, technology improves 1.5 times faster and carbon removal gradually increases. This is the most feasible scenario as the required changes are comparatively moderate despite assumptions in improvements to efficiency and carbon intensity over a long period of time. The main hurdle is that slower economic growth and improved technology can substantially reduce warming, but does not 'stop' it.

Scenario 2 requires significantly stronger intervention as GDP per capita growth is reduced to 60% of the baseline and technological improvement is doubled. Carbon removal also rises more rapidly so that net emissions approach zero, thus allowing temperature to stabilise just below 2°C. While this is mathematically achievable within the constraints of our model, it is significantly less feasible in practice as it depends on sooner and more significant reductions in emissions, along with more rapid developments to technology. This suggests that delaying action on the variables in practice increases the amount of carbon removal required over time.

Scenario 3 is the least realistically feasible, whereby GDP per capita growth is reduced to 50% of the original projection, technology improves 2.5 times more quickly and carbon removal exceeds 90 gigatons by the end of the century. These changes are necessary, as under the Rahmstorf sea level model, sea levels are predicted to continue to rise as long as temperatures remain above the pre-industrial baseline. Therefore, halting sea level rise requires temperatures to approach the pre-industrial baseline, which requires sustained net-negative emissions. While this outcome is mathematically feasible within our model, the scale of carbon removal required makes it very impractical and unrealistic given current technological conditions.

Population remains unchanged across all three scenarios thus meaning that stricter climate targets rely on reducing emissions through slower GDP growth, faster technological improvement and greater carbon removal as opposed to demographic change. However, if the population were to decline in the future, our model would predict lower annual emissions, assuming other factors remain unchanged including any exogenous events such as military conflict, pandemics or global financial crises that could affect mortality or birthrates. This would reduce cumulative emissions, thus lowering the projected temperature trajectories and sea level rise trajectories. However, since population and economic growth are modelled separately, our model does not account for how any substantial declines in population could affect economic activity or developments in technology.

Many limitations should be considered when interpreting the results produced by our model. The model simplifies complex climate and socioeconomics by using fixed functional forms for population, GDP growth and technological improvement, all whilst combining energy intensity and carbon intensity into a single technology term. The carbon removal term is also able to increase to the level required by each scenario without directly accounting for real world constraints such as costs, technology infrastructure and energy demands. Additionally the TCRE and Rahmstorf relationships provide simplified representations of temperature and sea level responses. Future research could address these limitations by applying more realistically feasible limits to carbon removal and by testing alternative population and economic pathways and incorporating more detailed climate and sea level feedbacks.

5 Conclusion

The model that we have created suggests that climate change can be stabilised. However more ambitious targets require much stronger intervention whereby stabilising temperature requires net emissions to approach net zero and reducing temperature requires sustained net-negative emissions. Scenario 1 is the most realistically feasible, and Scenario 2 requires more advancements in technology and carbon remove methods in order to happen. But Scenario 3, while it is mathematically possible, is not feasible in the real world due to its extreme carbon-removal requirements.

Our simplified model shows that technology and carbon removal must be complemented with rapid emissions reductions as opposed to outright replacing it. These targets are mathematically possible, but under current trends, are not yet practically feasible.

6 Bibliography

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A Appendix

A.1 Figures

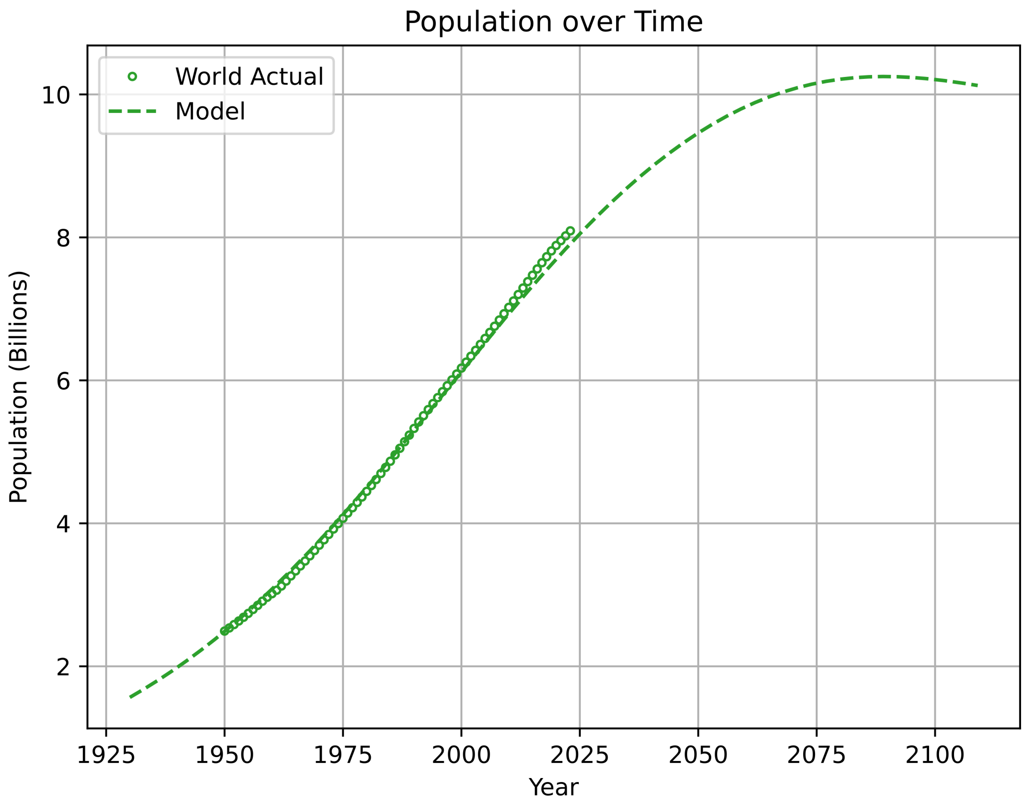

Figure 5: Population model compared to real world population, dataset from Our World in Data for years 1950–2023. mean percentage error = 0.0987%, percentage error stdev = 1.478%

Intuition behind the model: Many papers argue, and UN projections also expect that population will eventually peak towards the end of this century between 9 and 12 billion people, before declining slowly. The UN projections use more complex ideas to calculate these projections, taking into account decreasing fertility rates, increasing literacy rates, and economic development occurring in lesser developed countries. The population decline term amending our population model empirically captures exactly that, so that the population model is more accurate.

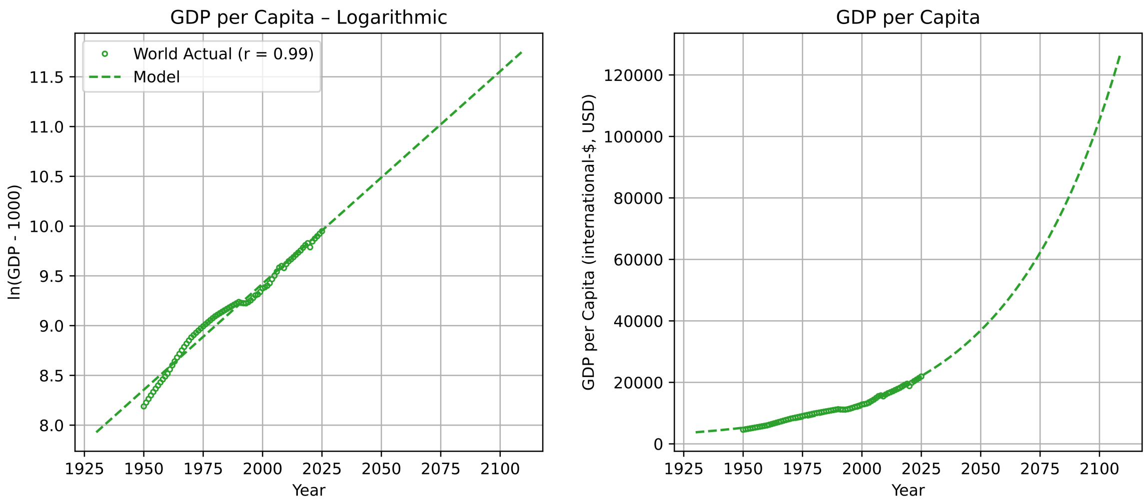

Figure 6: GDP per capita model compared to real world historical GDP, dataset from Our World in Data for years 1950–2025. mean percentage error = 0.1199%, percentage error stdev = 5.93%

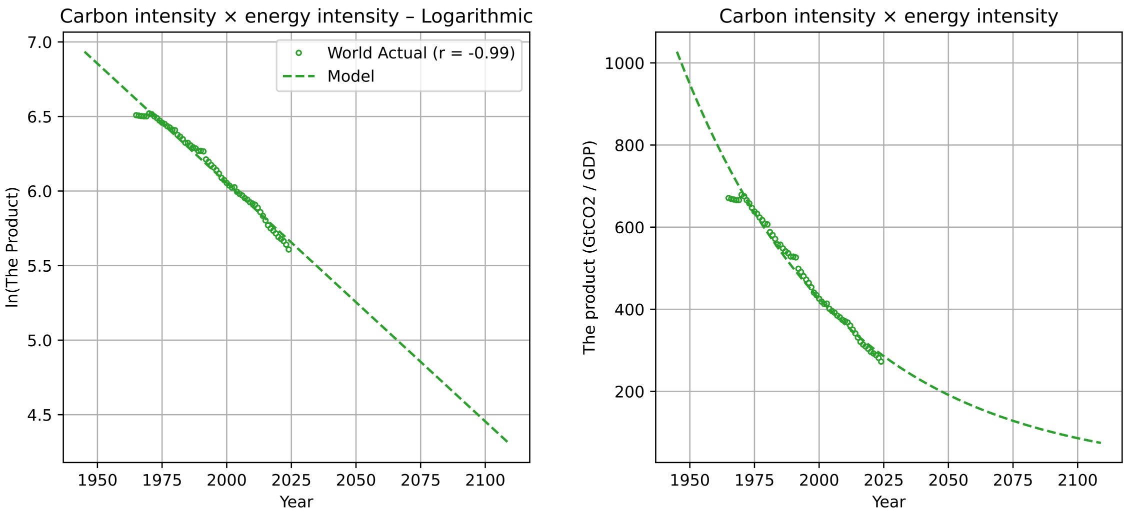

Figure 7: Technological inefficiency model compared to real world historical carbon intensity $\times$ energy intensity, dataset from Our World in Data for years 1965–2024. mean percentage error = 0.061%, percentage error stdev = 3.53%

The terms in both of these exponential functions were determined by linear regression on the logarithms of the values.

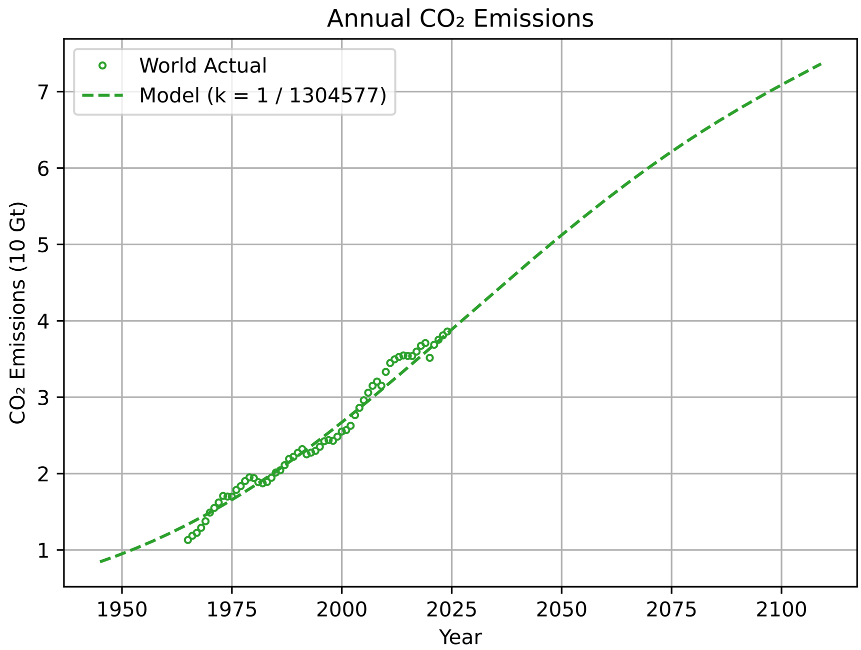

Figure 8: CO_2 model compared to real world historical CO_2 emissions, dataset from Our World in Data for years 1965–2024. mean percentage error $\approx 0\%$, percentage error stdev = 5.5359%

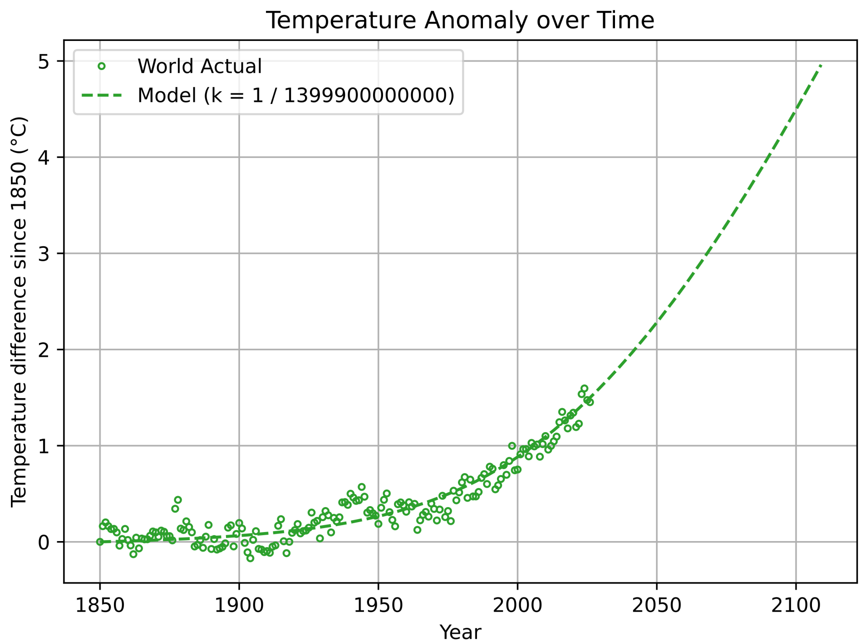

Figure 9: Temperature model compared to real world historical temperature anomaly, dataset from Our World in Data for years 1850–2026. mean error $\approx 0°C$, error stdev = 0.1259°C

These functions where calculated from Equation 3 and Equation 1, respectively.

Since these were both proportionalities, we determined the constants of proportionalities by selecting some denominator, then incrementing/decrementing it until the mean error was as low as possible. For these, the error was on the order of 2E-5 (2×10^{-5}).

While approaching scenarios 1 and 2, in order for the temperature difference to actually plateau at 4°C in 2100 or 2°C in 2050, the required CO_2 removal would need to nearly match CO_2 production within the span of a few months. We deemed this not feasible because of the rapid technological deployment and cost to reach multiple gigatonnes of CO_2 removal. So we opted to use a piecewise continuous removal function, which can be described as a ramping effort.

A.2 Our Code

We have included a link to our code used to model in our bibliography. But the link is also included here: `https://github.com/vedangpurohit21/modelling3041`.